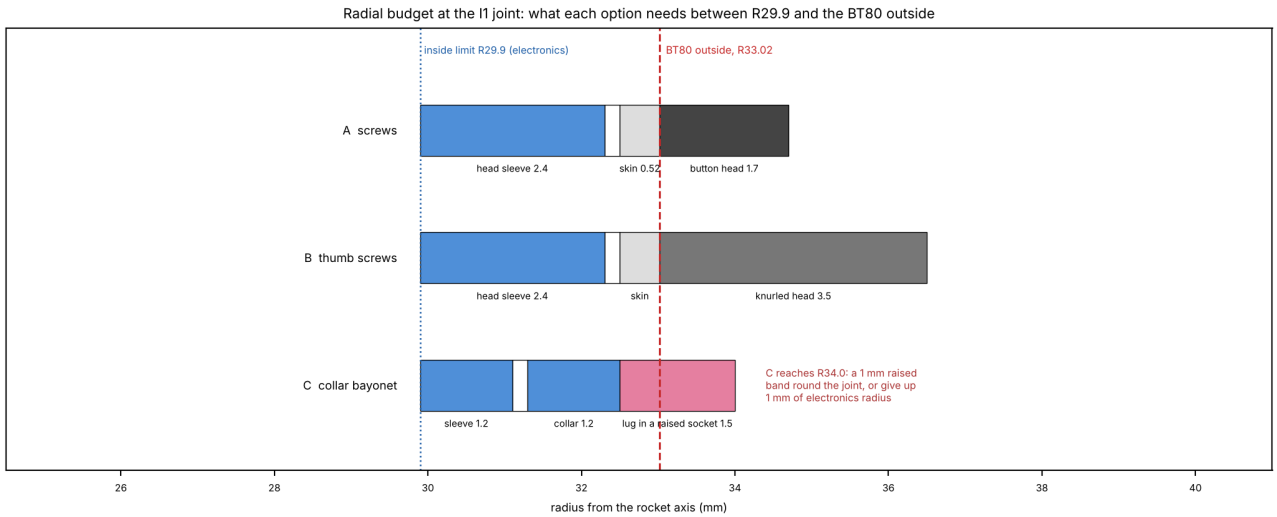
Choice 2: how the head and the module are fastened

The swap points are the joints I1 (head to module) and I2 (module to base). To swap the module you open both; to move the head to another brake you open I1 and the joint above it. There are three ways to make these joints. The drawings show why the third is awkward: the space between the electronics and the outside of the rocket is only **3.1 mm** in the radial direction. Concept numbers, not CAD.

1 The three options

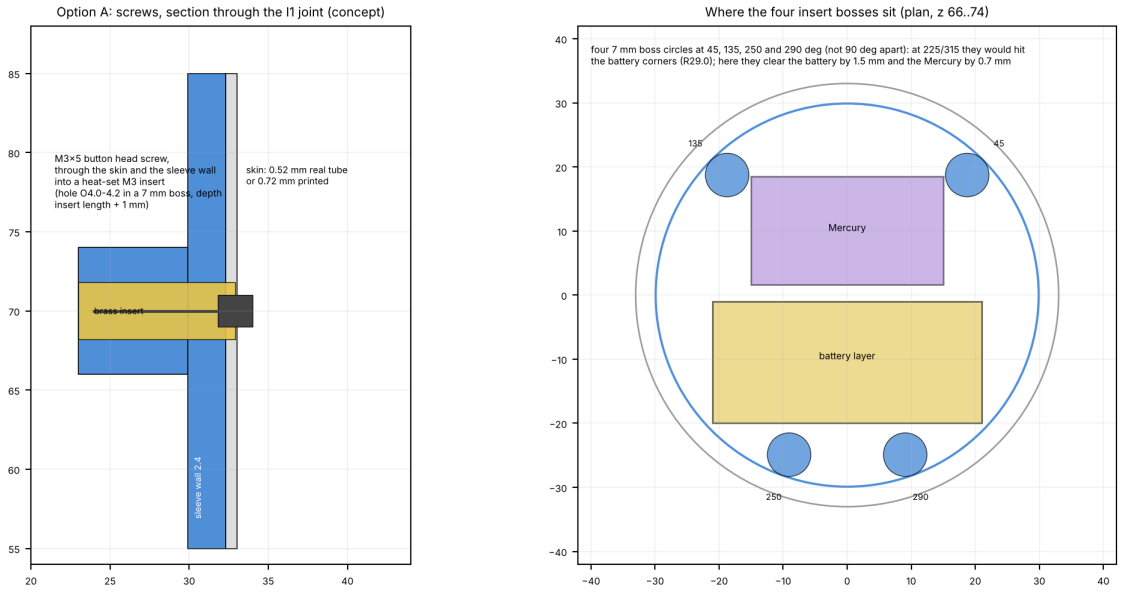
	A: screws (with heat-set inserts)	B: captive thumb screws	C: collar bayonet
What it is	Four M3×5 button-head screws, radial, through the skin and the head sleeve into brass inserts in four 7 mm bosses inside the sleeve. Same at I2.	The same four bosses and inserts, but with knurled M3 thumb screws that you turn by hand.	Three lugs on a ring (collar) that turns 24.5° in three L-slots in a raised socket. The collar turns; the head body does not.
Tools	2.0 mm hex driver	none	none (one safety screw or pin as a secondary lock)
Swap effort (estimate, not timed)	8 screws out and in: about 5 minutes	about 1 minute	about 1 minute
Strength	best: steel screws in brass inserts carry shear and tension; used in V4 already	about the same as A if the thumb screw is steel; the heads stand 3.5 mm proud	weakest: three 6 mm printed lugs in a 1.2 mm collar; it needs testing
Wear	printed threads strip after repeated swaps, so use heat-set inserts (the checklist's advice for parts assembled many times)	same as A (inserts)	lugs and slots wear; a detent bump wears
Surface	screw heads 1.7 mm proud	heads about 3.5 mm proud: a flush recess does not fit (the wall is 3.1 mm)	a 1 mm raised band round the joint (R34.0), or 1 mm of electronics radius given up
Main catch	the slowest to swap and screws can be lost	bigger heads in the airstream; hardware choice must be checked	the whole head cannot be twisted (see drawing): the rotor would move

2 What each one costs in the design

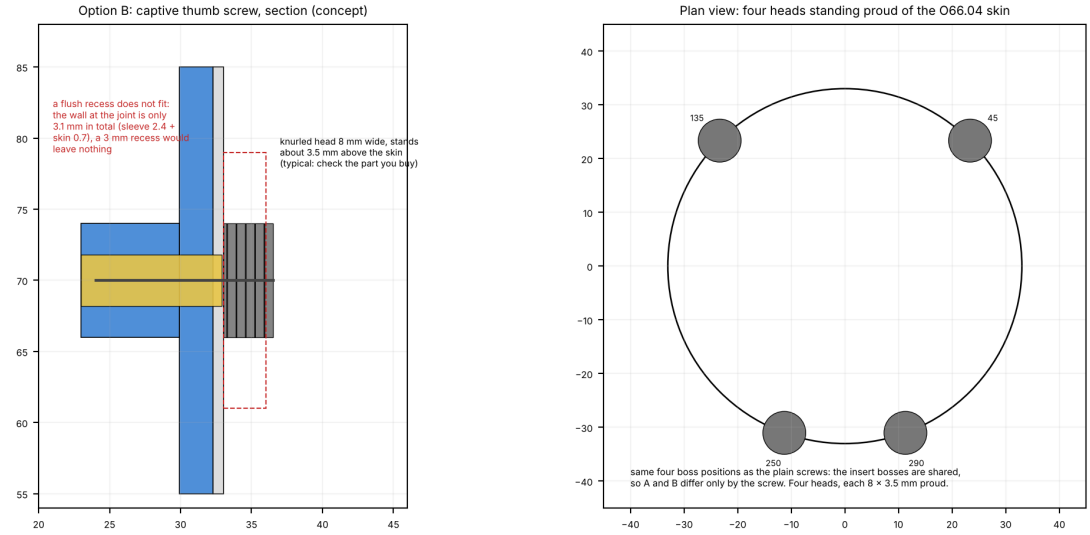


Radial budget at I1: the head sleeve, the skin and the fastener between R29.9 (the electronics' inside limit) and R33.02 (the BT80 outside).

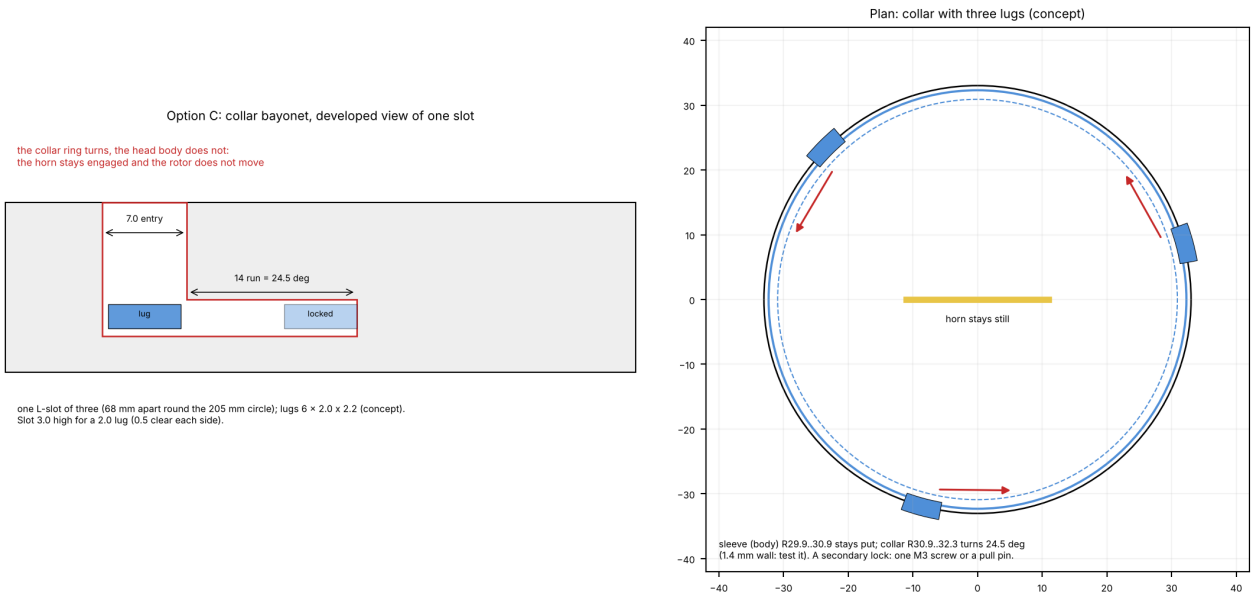
3 Drawings



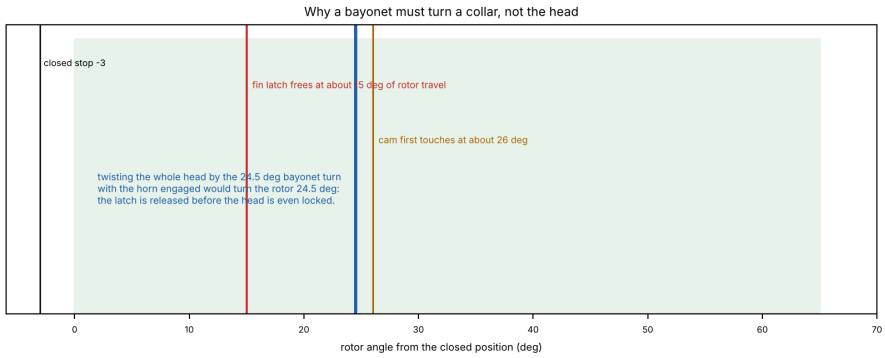
Option A: section through the joint (left) and the four insert bosses in plan (right). The bosses are at 45, 135, 250 and 290° so they clear the battery and the Mercury; at 225/315° they would hit the battery corners. The earlier concept sheet drew them 90° apart: this plan corrects it.



Option B: the same bosses with captive thumb screws; the heads stand proud because a recess does not fit.



Option C: one L-slot developed flat (left) and the collar with three lugs in plan (right).



A bayonet must turn a collar, not the head: the horn is engaged in the rotor, so turning the whole head would turn the rotor past the point where the fin latch lets go.

4 What I recommend

- **Start with Option A, screws into heat-set inserts.** It uses hardware V4 already has, it is the strongest, and the inserts survive many swaps. The four bosses go in the head sleeve at 45, 135, 250 and 290°.
- **Option B is a screw swap, not a redesign.** It uses the same bosses and inserts, so you can try thumb screws on the same parts. Check the real head height: if the heads stand 3.5 mm proud it is a trade of speed for a small obstacle in the airstream.
- **Leave Option C for later.** It needs a raised 1 mm band or less electronics radius, three 6 mm printed lugs in thin walls, and a collar that has to be tested; the drive coupling stays safe only if the collar, not the head, turns.
- **Decide where the rocket separates for recovery.** If the airbrake joints are where the airframe comes apart, they must carry the parachute shock, which favours screws. If the rocket separates elsewhere, the joints see only compression and handling loads.
- **First test (cheap):** print a 20 mm slice of the sleeve and skin with one boss, press in one insert, and screw it ten times. If the insert holds, build the full I1 sleeve.
- **Not verified:** every number is a concept dimension (3.1 mm radial space, boss sizes, lug sizes, swap times). The 24.5° turn of Option C and the latch-release angle of about 15° of rotor travel come from the V4 model; the rest is estimated.